

Supplementary File: Adaptive Data-Resilient Multi-Modal Hierarchical Multi-Label Book Genre Identification

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APPENDIX A DATASET DETAILS

To the best of our knowledge, there is hardly any publicly available dataset for hierarchical multi-label book genre prediction with multi-modal information. To address this gap, we constructed a dataset from scratch. Initially, we web-scraped book cover images, blurb texts, metadata (such as authors and publishers), and associated genres from [Goodreads](#). However, Goodreads lists ~ 2055 user-defined genres, many of which are redundant, anomalous, or highly subjective. Hence, genre labeling required standardization.

To address the complexity of multi-label genre annotation, we simplified the task of managing numerous interrelated genres by grouping them into broader, hierarchically structured parent categories. This approach enhanced consistency and interpretability in the labeling process. At Level-1, books were classified into *fiction* and *non-fiction* categories. At Level-2, each book was annotated with 1 to 6 sub-genre labels (refer to Table II in the main paper), with 29 labels defined for each top-level category. Eight annotators, in consultation with three linguistic experts, performed the genre assignments. To ensure labeling consistency, we measured inter-annotator agreement using Krippendorff’s alpha (α) [1], which achieved a score of $\alpha = 0.83$, indicating a high level of agreement and annotation reliability. The final dataset contains 11302 books, comprising 6704 fiction and 4598 non-fiction samples.

In the multi-modal setting, each book sample comprised a cover image, blurb text, and metadata. Additionally, we engaged Gemini Pro Vision [2] to perform optical character recognition (OCR) on the cover images to extract embedded textual content/ cover texts.

The dataset challenges, data augmentation details, with statistical information are provided in <https://github.com/Utsav30/IMAGINE>.

APPENDIX B ANALYSIS WITH DIFFERENT FEATURE EXTRACTOR

This section evaluates the performance of various feature extractors integrated into IMAGINE’s core architecture, excluding the SGM module.

A. Visual Feature Extractors

Fig. B.1(a) reports the performance of UM_v with different visual backbones. In Level-1 classification, Swin-B [16] achieved the highest accuracy ($\mathcal{A}$), while CLIP [15] yielded a slightly higher F1-score ($\mathcal{F}$) by improving true positive classification at the cost of increased false positives.

For Level-2 classification, Swin-B consistently outperformed others across all metrics in the *fiction*

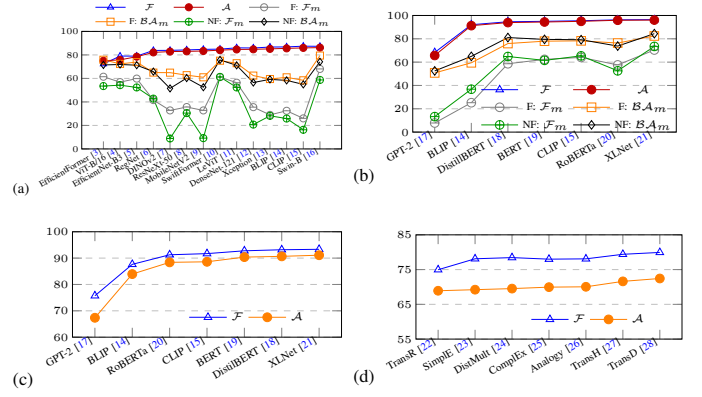


Fig. B.1: Comparison with (a) UM_v , (b) UM_d , (c) UM_c , and (d) UM_m

category. In *non-fiction*, SwiftFormer [10] showed marginally better results, making it the best-performing UM_v for that subset. However, Swin-B exhibited a substantial performance advantage in fiction and overall robustness across both categories. Therefore, Swin-B was selected as the visual feature extractor for its strong multi-scale representation capabilities and balanced performance.

B. Textual Feature Extractors

As shown in Fig. B.1(b), XLNet [21] consistently outperformed other models across all evaluation metrics at both levels for both fiction and non-fiction categories. Thus, XLNet is chosen as the textual feature extractor due to its strong overall performance and effective handling of long-range dependencies without relying on masked tokens. A similar trend was observed for textual feature extraction from cover text, where XLNet demonstrated superior Level-1 classification performance, as shown in the Fig. B.1(c).

C. Feature Extractors from Metadata

Fig. B.1(d) presents the Level-1 classification performance of UM_m on metadata. Among the evaluated knowledge graph embedding models, TransD [28] achieved the best performance, owing to its dynamic modeling of entities and relations. Consequently, TransD was selected as the metadata feature extractor.

APPENDIX C QUALITATIVE RESULTS

A. Qualitative Analysis: Heatmap Encoding

Table C.1 presents some data samples of our dataset, and provides a qualitative evaluation of IMAGINE performance through corresponding heatmap encodings. The grayscale heatmaps show the ground-truth and IMAGINE-predicted genre labels, with white representing hits and black indicating missed genres.

TABLE C.1: Qualitative assessment on performance of IMAGINE

(a) Data samples

(a) 9780060223595: Bread and Jam for Frances; Author: Russell Hoban, Lillian Hoban; Publisher: HarperCollins	
	The classic funny story about a stubborn little badger with very particular tastes in food. Frances is a fussy eater. In fact, the only thing she likes is bread and jam. She won't touch her squishy soft-boiled egg. She trades away her chicken-salad sandwich at lunch. She turns up her nose at boring veal cutlets. Unless her parents can come up with a plan, Frances just might go on eating bread and jam forever! Join the generations of families that have laughed along as Frances sings "Jam on biscuits, jam on bread, Jam is the thing I like most, Jam is sticky, jam is sweet, Jam is tasty, jam's a treat—raspberry, strawberry, gooseberry, I'm very FOND...OF...JAM!"
(b) 9782723423014: Sailor Moon, tome 14 : Le royaume Elusion; Author: Naoko Takeuchi; Publisher: GLÉNAT	
	A graphic novel offers the continuing adventures of Serena and her friends, who use the power of the moon and the planets, as the superhero Sailor Moon and her comrades, to battle the forces of evil.
(c) 9780895297273: Prescription for Nutritional Healing: A Practical A-Z Reference to Drug-Free Remedies Using Vitamins, Minerals, Herbs & Food Supplements; Author: James F. Balch, Phyllis A. Balch; Publisher: Avery	
	An alphabetical encyclopedia of nutritional treatments for a wide range of common ailments explains how to use vitamins, herbs, minerals, and food supplements to promote healing and wellness.
(d) 9780060172732: The Dog Who Rescues Cats: The True Story of Ginny; Author: Philip González, Leonore Fleischer; Publisher: HarperCollins	
	The true story of Ginny, an abused dog with an affinity for rescuing cats in trouble, describes Ginny's part in the saving of such handicapped felines as the one-eyed Revlon, deaf Madam, and the paralyzed Topsy.
(e) 9780671668365: The Magic of Conflict: Turning a Life of Work into a Work of Art; Author: Thomas F. Crum; Publisher: Touchstone	
	Gives techniques for transforming conflict into an opportunity for choice and change and how to grow beyond the striving competitiveness, and pressure of personal and professional life.
(f) 9780963235930: Beyond Backpacking: Ray Jardines Guide to Lightweight Hiking; Author: Ray Jardine; Publisher: AdventureLore Press	
	Beyond Backpacking has 517 pages of "practical methods for all who love the out-of-doors". Chapters include Equipment, Hiking Considerations, Obstacles, Longer Journeys and Back to Basics.
(g) 9780380773527: Dreaming of You; Author: Lisa Kleypas; Publisher: Avon	
	She stood at danger's threshold—then love beckoned her in. A prim, well-bred gentlewoman, Sara Fielding is a writer who puts pen to paper to create dreams. But now curiosity is luring her from the shelter of her country cottage into the dangerous world of Derek Craven—handsome, tough, and tenacious—and the most exciting man Sara has ever met. Derek rose from poverty to become the wealthy lord of London's most exclusive gambling house. And now duty demands that he allow Sara Fielding to enter his perilous realm of ever-shifting fortunes—with her impeccable manners and her infuriating innocence. But ...
(h) 9780679602613: The Soul of a New Machine; Author: Tracy Kidder; Publisher: Modern Library	
	Winner of the Pulitzer Prize and the American Book Award, The Soul of a New Machine was a bestseller on its first publication in 1981. With the touch of an expert thriller writer, Tracy Kidder recounts the feverish efforts of a team of Data General researchers to create a new 32-bit superminicomputer. A compelling account of individual sacrifice and human ingenuity, The Soul of a New Machine endures as the classic chronicle of the computer age and the masterminds behind its technological advances. "A superb book," said Robert Pirsig, author of Zen and the Art of Motorcycle Maintenance ...
(i) 9780803716988: Library Lil; Author: Suzanne Williams; Publisher: Dial	
	A formidable librarian makes readers not only out of the once resistant residents of her small town, but out of a tough-talking, television-watching motorcycle gang as well.
(j) 9780130255792: Understanding Movies; Author: Louis D. Giannetti; Publisher: Prentice Hall	
	Designed to help movie watchers analyze films with precision and technical sophistication, this book focuses on formalism how the forms of the film (e.g., camera work, editing, photography, etc.) create meaning. It sheds light on how television and movies communicate, and the complex network of language systems they use. Chapter topics cover recent developments from all aspects of cinema, contemporary films, personalities in the field, photography, movement, editing, sound, acting, drama, story writing, and theory. For movie critics and fans alike.

(b) Ground-truth and IMAGINE-predicted confidence score heatmap encodings in gray color code

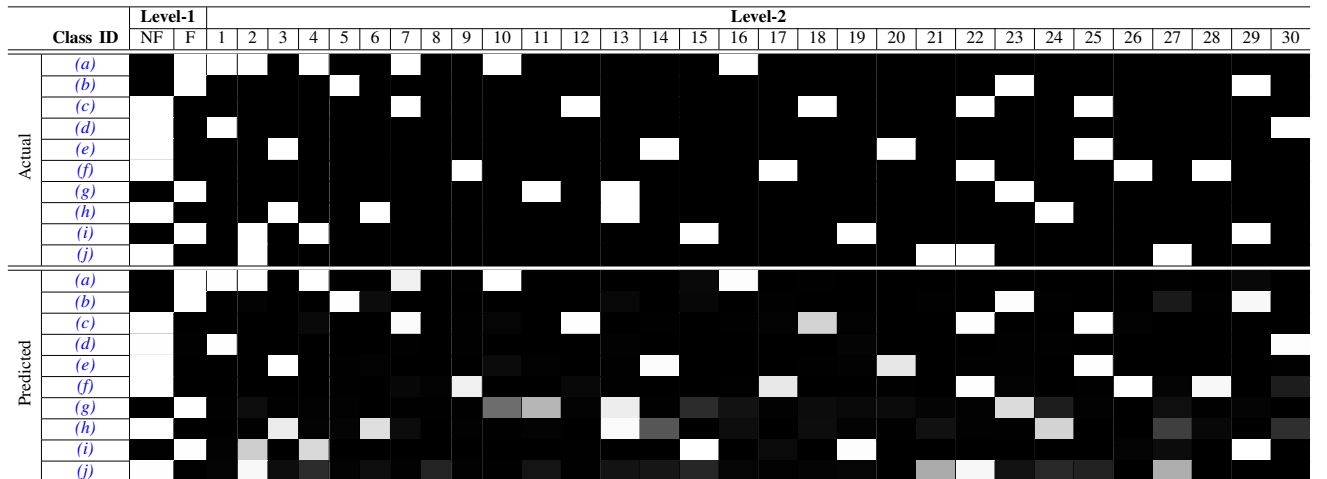


TABLE C.2: Misprediction due to insufficient information

(a) 9780760705902: The Awakening; Author: Kate Chopin; Publisher: Barnes & Noble Books		
	Blurb	The Awakening
	Actual Genre	(Fiction, {History, Humanities, Literature, Teen & Young Adult})
	Predicted Genre	(Fiction, {Arts & Photography, Environment & Plant, Literature})
(b) 9780826400475: Pedagogy of the oppressed; Author: Paulo Freire; Publisher: Continuum Publishing Co. (NY)		
	Blurb	Signed copy by the author
	Actual Genre	(Non-fiction, {Humanities})
	Predicted Genre	(Non-fiction, {Science & Math})
(c) 9788420478722: La caverna de las ideas; Author: José Carlos Somoza; Publisher: Alfaguara		
	Blurb	Book by Jose Carlos Somoza
	Actual Genre	(Fiction, {History, Humanities, Literature, Mystery & Thriller & Suspense & Horror & Adventure})
	Predicted Genre	(Fiction, {Arts & Photography, Literature, Reference & Language, Sci-Fi & Fantasy})
(d) 9781571740601: Gooberz; Author: Linda Goodman; Publisher: Hampton Roads Publishing		
	Blurb	Paperback
	Actual Genre	(Fiction, {Literature, Mythology & Religion & Spirituality})
	Predicted Genre	(Fiction, {Arts & Photography, Literature, Sci-Fi & Fantasy})

B. Qualitative Analysis for Misprediction

Our analysis of mispredictions suggests that a key contributing factor is the lack of sufficient information in certain samples. In some cases, even human evaluators struggle to accurately determine the genre due to ambiguous or limited content. Table C.2 showcases examples where IMAGINE fails to predict the correct genre. Notably, some samples are prone to misinterpretation due to inadequate blurb content. Fig. C.1 presents cover images containing misleading visual cues that influenced IMAGINE’s genre prediction. In these cases, the model relied on visual cues, resulting in predicted genres consistent with the cover design but inconsistent with the actual content, leading to misprediction.

Based on these findings, we recommend enhancing the quality of blurbs with more detailed descriptions and redesigning cover pages to more clearly reflect the book’s true genre. These improvements can reduce genre misclassification, attract the appropriate audience, and improve overall content alignment with reader expectations.

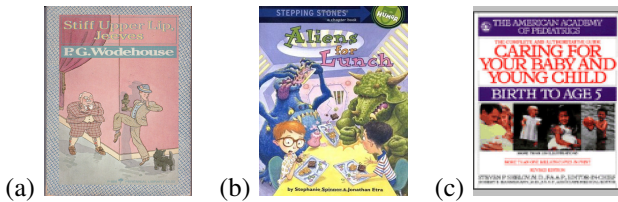


Fig. C.1: Mispredictions: IMAGINE classified (a)-(b) as comics & graphics, and (c) as medical.

C. Qualitative and Quantitative Analysis for Genre Indistinguishability

In multi-label classification, especially with imbalanced datasets, label co-occurrence can significantly affect prediction accuracy. This becomes particularly problematic when a dominant genre frequently overlaps with a less represented one, leading the model to underrepresent the minority due

to skewed training distributions. Such patterns can result in false positives or false negatives, and obscure the actual genre structure. The interaction between *sci-fi & fantasy* and *press & media* in *fiction* category in our dataset exemplifies this issue, as illustrated below.

The training set contains 396 samples labeled as *press & media* and 3980 as *sci-fi & fantasy*, with only 264 samples overlapping between the two. This means that 93% of the *sci-fi & fantasy* training samples do not belong to *press & media*, while only 7% are shared. Conversely, 67% of the *press & media* samples also belong to *sci-fi & fantasy*, leaving just 33% exclusive to *press & media*. This co-occurrence imbalance leads to two key observations:

Case-(i): Because the vast majority (93%) of *sci-fi & fantasy* samples are not associated with *press & media*, the model tends to predict only *sci-fi & fantasy* for test instances that actually belong to both categories, leading to false negatives for *press & media* (refer to Table C.3:(a), (b)).

Case-(ii): Since only one-third of *press & media* training samples are exclusive, the model often overgeneralizes, predicting *sci-fi & fantasy* for test samples that are only labeled *press & media*, resulting in false positives for *sci-fi & fantasy* (refer to Table C.3:(c)).

These patterns highlight the challenges of genre prediction in multi-label settings with skewed co-occurrence distributions. More details of the qualitative and quantitative analysis of genre indistinguishability are provided in <https://github.com/Utsav30/IMAGINE>.

D. Qualitative Analysis of Model Limitation

Another significant reason for IMAGINE’s mispredictions is the inherent limitations of the model. From Table C.4, we observe genre misprediction in Level-2 classification due to misclassification in Level-1 classification for some samples.

REFERENCES

- [1] K. Krippendorff, “Estimating the reliability, systematic error and random error of interval data,” *Educational and Psychological Measurement*, vol. 30, no. 1, pp. 61–70, 1970.

TABLE C.3: Examples of misprediction due to genre indistinguishability

(a) 9780671501075 : Dragon's Honor (Star Trek: The Next Generation, No. 38); Author : KJ. Johnson, Greg Cox; Publisher : Star Trek		
	Blurb	Isolated for centuries, the exotic Dragon Empire is finally ready to join the United Federation of Planets. But first the emperor's eldest son must marry the only daughter of his oldest enemy, bringing to an end decades of civil war. Without the wedding, there can be no peace – and no treaty with the Federation. As honored guests of the Dragon Empire, Captain Picard and the crew of the "Starship Enterprise" TM must ensure that the royal wedding occurs on schedule, despite the Empire's complicated and difficult codes of honor. And Dr. Beverly Crusher finds her loyalties torn when she wins the confidence of the unusually reluctant bride-to-be. More than just of the Federation, Picard a treaty is at stake, for a vicious race of alien conquerors will stop at nothing, from assassination to invasion, to keep the Empire out must use all his skills to save the Empire, and preserve the Dragon's Honor.
	Actual Genre	(Fiction, (Press & Media, Sci-Fi & Fantasy))
	Predicted Genre	(Fiction, (Sci-Fi & Fantasy))
(b) 9781569711293 : Star Wars: Battle of the Bounty Hunters; Author : Ryder Windham, Christopher Moeller; Publisher : Dark Horse		
	Blurb	Battle of the Bounty Hunters was a pop-up comic and part of the Star Wars: Shadows of the Empire multimedia campaign. It showed the space battle between Boba Fett and IG-88D.
	Actual Genre	(Fiction, (Comics & Graphic, Press & Media, Sci-Fi & Fantasy))
	Predicted Genre	(Fiction, (Comics & Graphic, Sci-Fi & Fantasy))
(c) 9780688142964 : Simple Simon; Author : Ryne Douglas Pearson; Publisher : William Morrow & Co		
	Blurb	A ruthless assassin has been sent to kill Simon Lynch, Autistic teenager and mathematical genius who breaks the National Security Agency's latest code. But will she succeed? Follow FBI agent Art Jefferson in his fight to protect Simon until the final riveting moments. Guaranteed to keep you guessing. Bruce Willis will star in the Ron Howard Film based on this book.
	Actual Genre	(Fiction, (Medical, Mystery & Thriller & Suspense & Horror & Adventure, Press & Media))
	Predicted Genre	(Fiction, (Medical, Mystery & Thriller & Suspense & Horror & Adventure, Press & Media, Sci-Fi & Fantasy))

False positives are marked in red, and False negatives are marked in blue

TABLE C.4: Misprediction due to Level-1 classification

(a) 9780312266820 : Dead Egotistical Morons; Author : Mark Richard Zubro; Publisher : St. Martin's Press		
	Blurb	Boys4U is the world's most popular singing group - at least among teenaged girls - and they have closed out their sold-out world tour with a series of shows in Chicago's brand new arena. The premier group of the inexplicably popular "boy band" trend, they've just finished their very last concert of the tour. While hundreds of tour members, well-wishers, executives, and various hangers on wait to celebrate another wildly successful tour, the lead singer is found murdered - shot in the back of the head at close range - in the shower all times, the only other people back there were the other members of the band, and none of the dozens of people in the next room report having heard a shot. While the international press is engaged of the backstage dressing area. To make matters more distressing, the crime itself was almost impossible - there was tight security on the shower area at in an unprecedented feeding frenzy over the sensationalistic murder. Chicago Police Detectives Paul Turner and his partner Buck Fenwick have pulled the unenviable task of investigating the murder But even the initial appearances are deceiving and as they dig deeper into the case, they uncover more disturbing truths beneath the wholesome façade of Boys4U. Now they have untangle an increasingly complex web if they are to stop a determined killer before more victims are claimed.
	Actual Genre	(Fiction, (Fashion & Lifestyle, Mystery & Thriller & Suspense & Horror & Adventure))
	Predicted Genre	(Non-fiction, (Arts & Photography))
(b) 9780312979256 : Mafia Marriage; Author : Rosalie Bonanno, Beverly Donofrio; Publisher : St. Martin's Paperbacks		
	Blurb	An Unforgettable Look Inside the Godfather's Own House She Was A Profaci. He Was A Bonanno Rosalie Profaci was a Mafia princess. Salvatore "Bill" Bonanno, oldest son of Mafia Don Joe Bonanno, the real-life model for The Godfather, was organized crime's crown prince. And Bill, deeply involved in his father's "business" of mob schemes thought pretty Rosalie knew what it meant to be a "Mafia wife." But the convent-raised, deeply devout Rosalie, whose innocence was protected by her doting father, had no idea... Their Marriage United Two Mafia Dynasties... Mafia Marriage is Rosalie Bonanno's intimate account of life inside the secretive world of the Mafia. Naming names and providing shocking details, she writes about the wild spending sprees, the mysterious absences of her husband, the other women in his- their running from the law, the abductions, and shootings. Above all, Rosalie reveals the passion that kept her virtually a prisoner to love... and her heartbreaking journey of discovering the truth and trying to break free.
	Actual Genre	(Non-fiction, (Biographies & Memoir, Mystery & Thriller & Suspense & Horror & Adventure))
	Predicted Genre	(Fiction, (History, Romance, Teen & Young Adult))

- [2] R. Anil *et al.*, "Gemini: A family of highly capable multimodal models," *arXiv:2312.11805*, 2023.
- [3] Y. Li *et al.*, "Efficientformer: Vision transformers at mobilenet speed," *NeurIPS*, vol. 35, pp. 12 934–12 949, 2022.
- [4] A. Dosovitskiy *et al.*, "An image is worth 16x16 words: Transformers for image recognition at scale," in *ICLR*, 2021.
- [5] M. Tan *et al.*, "EfficientNet: Rethinking model scaling for convolutional neural networks," in *ICML*, vol. 97, 2019, pp. 6105–6114.
- [6] J. Xu *et al.*, "RegNet: Self-regulated network for image classification," *IEEE TNNLS*, vol. 34, no. 11, pp. 9562–9567, 2023.
- [7] M. Oquab *et al.*, "DINOv2: Learning robust visual features without supervision," *TMLR*, 2024.
- [8] S. Xie *et al.*, "Aggregated residual transformations for deep neural networks," in *CVPR*, 2017.
- [9] M. Sandler *et al.*, "Mobilenetv2: Inverted residuals and linear bottlenecks," in *CVPR*, 2018, pp. 4510–4520.
- [10] A. Shaker *et al.*, "Swiftformer: Efficient additive attention for transformer-based real-time mobile vision applications," in *ICCV*, 2023, pp. 17 425–17 436.
- [11] B. Graham *et al.*, "LeViT: a vision transformer in convnet's clothing for faster inference," in *ICCV*, 2021, pp. 12 259–12 269.
- [12] G. Huang *et al.*, "Densely connected convolutional networks," in *CVPR*, 2017, pp. 4700–4708.
- [13] F. Chollet, "Xception: Deep learning with depthwise separable convolutions," in *CVPR*, 2017, pp. 1251–1258.
- [14] J. Li *et al.*, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *ICML*, 2022, pp. 12 888–12 900.
- [15] A. Radford *et al.*, "Learning transferable visual models from natural language supervision," in *ICML*, 2021, pp. 8748–8763.
- [16] Z. Liu *et al.*, "Swin transformer: Hierarchical vision transformer using shifted windows," in *ICCV*, 2021, pp. 10 012–10 022.
- [17] A. Radford *et al.*, "Language models are unsupervised multitask learners," *OpenAI blog*, vol. 1, no. 8, p. 9, 2019.
- [18] V. Sanh *et al.*, "Distilbert, a distilled version of bert: Smaller, faster, cheaper and lighter," 2019.
- [19] J. Devlin *et al.*, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *NAACL-HLT, 2019*, Jun. 2019, pp. 4171–4186.
- [20] Y. Liu, "Roberta: A robustly optimized bert pretraining approach," 2019.
- [21] Z. Yang *et al.*, "XLNet: Generalized autoregressive pretraining for language understanding," in *NeurIPS*, 2019.
- [22] Y. Lin *et al.*, "Learning entity and relation embeddings for knowledge graph completion," in *AAAI*, vol. 29, no. 1, 2015.
- [23] S. M. Kazemi *et al.*, "Simple embedding for link prediction in knowledge graphs," in *NeurIPS*, 2018, p. 4289–4300.
- [24] B. Yang *et al.*, "Embedding entities and relations for learning and inference in knowledge bases," *arXiv preprint arXiv:1412.6575*, 2014.
- [25] T. Trouillon *et al.*, "Complex embeddings for simple link prediction," in *ICML*, 2016, pp. 2071–2080.
- [26] H. Liu *et al.*, "Analogical inference for multi-relational embeddings," in *ICML*, 2017, pp. 2168–2178.
- [27] Z. Wang *et al.*, "Knowledge graph embedding by translating on hyperplanes," in *AAAI*, vol. 28, no. 1, 2014.
- [28] G. Ji *et al.*, "Knowledge graph embedding via dynamic mapping matrix," in *ACL-IJCNLP*, 2015, pp. 687–696.